OTTO VON GUERICKE UNIVERSITÄT MAGDEBURG NAT FAKULTÄT FÜR NATURWISSENSCHAFTEN

Materials Science: Team Project

Subject:

Impurity doping versus polarization doping in nitride materials

Description:
Impurity doping involves introducing foreign atoms into the nitride material. In contrast, polarization doping creates an effective charge carrier concentration by exploiting the inherent piezoelectric and spontaneous polarisation in graded alloy layers, such as AlGaN. Polarisation doping offers significant advantages in certain applications, such as higher conductivity and improved device performance (e.g. increased LED efficiency), and can eliminate the need for conventional p-type doping, which is difficult to implement.

Work packages:

- **Literature Review**: The main focus of the project is conducting a comprehensive literature review on the chosen topic. Students should gather relevant papers and research articles.
- **Problem Identification**: Identify key problems or challenges within the chosen topic. This should involve understanding the current state of research and the unresolved issues in the field.
- **Methodological Approaches**: Students should investigate various approaches and methodologies used in the literature to address the identified problems.
- **Comparison of Approaches**: Compare and contrast the different methods found in the literature, discussing their strengths, weaknesses, and applicability to the identified problems.
- **Project Motivation**: At the end of the project, students should write a one-page summary introducing the project, explaining the motivation behind the topic and its relevance.
- **Visual Elements**: Students may include images or other visual materials to support their motivation page.
- **Presentation**: Prepare a presentation summarizing the findings and progress made during the project. This should include insights from the literature research, proposed models, and solutions explored.
- **Discussion**: After the presentation, a discussion should follow where students can explain their work, address questions, and evaluate the effectiveness of the approaches explored.